

A Literal Signed Reduced-Residue Operator and Phase-Character Firewall

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August 26, 2026

Abstract

TPC-261 identified a strict fixed-power gap of $1/400$ between the current V59 baseline and target, while TPC-260 identified a four-packet reassembly datum. We now freeze the literal reduced-residue signed operator at each additive phase. With the actual residue synthesis $S_{q,v}$, unit mask P_q , and outer prime weight retained, the operator is $J_{q,v} = S_{q,v}^* C_q S_{q,v} - (q-2)(q-1)^{-1} P_q$. We prove its exact quadratic remainder identity and the associated four-phase polarization. We also prove that aggregate packet mode zero and the nontrivial polarization characters are distinct. An exact finite operator-image certificate on $\{5, 7, 11, 13\}$ gives identical diagonals with mode-zero energies $16 \|Y\|^2$ and zero. This is a finite structural obstruction, not a growing-shell counterexample or an estimate for the source-frozen packets.

1 Scope and motivation

The Route-B object is a signed coupling on one V59 clock. Earlier releases constructed the literal source operator, a four-block Haar frame, a source-frozen null direction, and an exact endpoint budget [1, 2, 3]. The remaining question is what exact quantity must be controlled before a packet diagonal estimate can be reassembled into the physical scalar?

The unit-class centering matrix is a constant-character fiber, not by itself the additive variable $v = 0$ of the complete V59 integral. We retain this distinction throughout. The finite certificate audits $v = 0$, while the operator identity is phase-by-phase and the growing integral remains open.

1. identify the exact unit-class projection and signed remainder operator;
2. prove the mode-zero/cross-Gram identity and phase-character separation;
3. translate the signed expression into the strict $1/400$ criterion; and
4. certify a literal finite operator-image adversary with rational arithmetic and an independent checker.

2 The literal reduced-residue fiber

For an odd prime q , write $m_q = q - 1$ and let $\mathbf{1}_q \in \mathbb{R}^{m_q}$ be the all-ones vector. Define

$$C_q = I_{m_q} - \frac{1}{m_q} \mathbf{1}_q \mathbf{1}_q^\top. \quad (1)$$

The rows correspond to the nonzero residue classes; no ordinary residue class is inserted.

Lemma 1 (unit-class projection). *For every odd prime q , C_q is symmetric and idempotent, is positive semidefinite, has rank $q - 2$, and has kernel $\mathbb{R}\mathbf{1}_q$.*

Proof. Since $\mathbf{1}_q^\top \mathbf{1}_q = m_q$, direct multiplication gives $C_q^2 = C_q$. For $v \in \mathbb{R}^{m_q}$,

$$v^\top C_q v = \|v\|^2 - \frac{|\mathbf{1}_q^\top v|^2}{m_q} = \frac{1}{m_q} \sum_{r < s} (v_r - v_s)^2 \geq 0. \quad (2)$$

The constant line is the kernel and its orthogonal complement is fixed, so the rank is $m_q - 1 = q - 2$. $\square$

For a finite set $\mathcal{Q}$ of odd primes, a packet source is $a = (a_q)_{q \in \mathcal{Q}}$, $a_q \in \mathbb{R}^{m_q}$. Its zero-fiber output is

$$Y(a)_q = C_q a_q, \quad \langle Y(a), Y(b) \rangle_{\mathcal{Q}} := \sum_{q \in \mathcal{Q}} q \langle C_q a_q, C_q b_q \rangle. \quad (3)$$

The prime weight is inside the inner product, so no unrecorded normalization change is made.

2.1 The signed remainder operator

Let I be a finite interval and $v \in \mathbb{R}$. For a coefficient vector $a \in \mathbb{C}^I$, define the physical residue synthesis

$$(S_{q,v} a)_r = \sum_{\substack{n \in I \\ n \equiv r \pmod{q}}} a_n e(vn/H), \quad r \in \mathbb{F}_q^\times. \quad (4)$$

Let P_q be the diagonal projection onto q -units. The signed reduced-residue remainder is represented exactly by

$$J_{q,v} = S_{q,v}^* C_q S_{q,v} - \frac{q-2}{q-1} P_q. \quad (5)$$

The first term is positive semidefinite, but $J_{q,v}$ is a signed Hermitian operator. Direct expansion gives

$$V_q^\times(a; v) - D_q^\times(a; v) = \langle a, J_{q,v} a \rangle, \quad V_q^\times(a; v) = \|C_q S_{q,v} a\|^2, \quad D_q^\times(a; v) = \frac{q-2}{q-1} \|P_q a\|^2. \quad (6)$$

Consequently the common-clock operator

$$\mathcal{J}_x = \int_{\mathbb{R}} \psi_+(v) \sum_{q \in \mathcal{Q}_x} q J_{q,v} dv \quad (7)$$

satisfies the exact V59 polarization identity

$$\mathfrak{C}_x = \frac{1}{4} \sum_{j=0}^3 \mathfrak{i}^j \langle a^{(j)}, \mathcal{J}_x a^{(j)} \rangle, \quad a^{(j)} = \beta + \mathfrak{i}^j w. \quad (8)$$

This is an exact factorization, not an estimate.

3 Mode zero is a signed cross-Gram

Let Y_0, Y_1, Y_2, Y_3 be four vectors in (3). Put

$$\Gamma_{jk} = \langle Y_j, Y_k \rangle_{\mathcal{Q}}, \quad D = \sum_{j=0}^3 \Gamma_{jj}, \quad R = \sum_{0 \leq j < k \leq 3} \operatorname{Re} \Gamma_{jk}. \quad (9)$$

Theorem 1 (literal cross-Gram mode-zero identity). *With*

$$\hat{Y}_k = \frac{1}{2} \sum_{j=0}^3 i^{-jk} Y_j, \quad 0 \leq k \leq 3, \quad (10)$$

one has

$$\|Y_0 + Y_1 + Y_2 + Y_3\|_{\mathcal{Q}}^2 = D + 2R, \quad (11)$$

$$\sum_{k=0}^3 \|\hat{Y}_k\|_{\mathcal{Q}}^2 = D, \quad (12)$$

$$\|Y_0 + Y_1 + Y_2 + Y_3\|_{\mathcal{Q}}^2 = 4 \|\hat{Y}_0\|_{\mathcal{Q}}^2. \quad (13)$$

Thus diagonal mass alone does not determine mode zero.

Proof. Expanding the first left side gives the diagonal terms and each off-diagonal term with its conjugate, hence $D + 2R$. Orthogonality of the four roots of unity gives Parseval for (10); setting $k = 0$ gives the last identity. $\square$

3.1 The phase-character firewall

The packet index has its own Fourier characters. For two vectors X, Y write

$$E_j = \|X + i^j Y\|^2, \quad F_k = \frac{1}{4} \sum_{j=0}^3 i^{kj} E_j. \quad (14)$$

With the conjugate-linear-first-slot convention, direct expansion yields

$$F_0 = \|X\|^2 + \|Y\|^2, \quad F_1 = \langle Y, X \rangle, \quad F_2 = 0, \quad F_3 = \langle X, Y \rangle. \quad (15)$$

Thus aggregate packet mode zero and the nontrivial character selected by the V59 polarized scalar are different observables. A theorem must name the character it estimates; a mode-zero bound cannot be silently substituted for the F_1 or F_3 coupling.

The identity is compatible with the V59 four-packet construction: whenever the packet outputs are produced by one literal linear map, every Γ_{jk} is a sum over the same primes and reduced-residue unit masks. The theorem does not replace the smooth kernel or prove a favorable sign.

Corollary 1 (endpoint translation). *The current V59 ledger is*

$$E_0 = \frac{5}{3} = \frac{2000}{1200}, \quad E_* = \frac{1997}{1200}, \quad E_0 - E_* = \frac{1}{400}. \quad (16)$$

If $D + 2R$ has effective fixed-power saving σ after all paid losses, the finite-lane endpoint compiler closes when $\sigma > 1/400$.

Proof. This is the exact exponent substitution into TPC-261's strict finite-lane compiler; strictness absorbs the arbitrary x^ε loss. $\square$

Table 1: Exact four-packet fiber audit.

Family	diagonal per packet	R	mode-zero energy
Aligned	15/4	45/2	60
Alternating	15/4	-15/2	0
Mixed (+, +, -, -)	15/4	-15/2	0

4 A finite literal operator-image adversary

Take the actual finite prime shell

$$\mathcal{Q} = \{5, 7, 11, 13\}. \quad (17)$$

Let e be the first coordinate in the $q = 5$ fiber and zero elsewhere, and let $Y = Y(e)$. Equation (2) gives $Y \neq 0$, and

$$\|Y\|_{\mathcal{Q}}^2 = 5 \left(\left(\frac{3}{4} \right)^2 + 3 \left(\frac{1}{4} \right)^2 \right) = \frac{15}{4}. \quad (18)$$

Define two families in the same literal operator image:

$$Y_j^+ = Y, \quad Y_j^- = (-1)^j Y. \quad (19)$$

Every packet in both families has diagonal 15/4, and both have total packet energy 15, but

$$\left\| \sum_j Y_j^+ \right\|^2 = 60, \quad \left\| \sum_j Y_j^- \right\|^2 = 0. \quad (20)$$

The corresponding R 's are 45/2 and -15/2. The difference is entirely in the signed cross-Gram, not in the diagonal.

This is a finite structural adversary. Its source probes are freely chosen and are not identified with $\beta + i^j w$ on the growing V59 interval. It refutes only the shortcut that literal diagonal and PSD information imply mode-zero saving. It does not refute a future theorem using the arithmetic of the actual source packets.

5 Route status and conclusion

TPC-262 pays one structural interface: the literal signed reduced-residue operator, its deleted diagonal, and its packet-index phase typing are now written exactly on one common finite object. It also identifies the exact failure mode of a diagonal-only argument after actual prime unit matrices are retained. The aggregate mode-zero identity is useful, but it is not silently identified with the nontrivial V59 polarization character.

The next experiment is a census of cross-Gram entries on a growing prime shell, restoring the smooth kernel, deleted diagonal, unit masks, and the source-frozen β, w inputs. A positive result must prove negative signed cross-Gram cancellation with effective credit strictly larger than 1/400. A finite census may instead reveal another obstruction; neither outcome is an asymptotic theorem by itself.

The named Session Route-A/Route-B evaluator files are absent from this checkout. The proof package, theorem ledger, exact certificate, bridge checker, and `AGENTS.md` are the available fail-closed authority.

References

- [1] L. Wang, "Literal V59 source-operator attachment," TPC-247 project artifact, 2026.
- [2] L. Wang, "Null-compatible four-packet residual reassembly," TPC-260 project artifact, 2026.
- [3] L. Wang, "A strict endpoint-budget compiler," TPC-261 project artifact, 2026.

Table 2: Claim firewall.

Item	Status
Unit-class projection and signed remainder operator	proved exactly
Mode-zero DFT and cross-Gram ledger	proved exactly
Phase-character separation	proved exactly
Endpoint threshold	proved exactly, conditional on lane bounds
Finite operator-image witness	numerically certified structural
Growing β, w cross-Gram estimate	open
Arithmetic L^2 , full Gate B, twin primes	none / open